

Numerical simulations of impinging jets with application to downbursts

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Abstract

Numerical simulations are employed to investigate the macro-flow dynamics and scale (Reynolds number) dependency of impinging jets with applications to downburst related high intensity winds. The flow is quasi-periodic with vortex rings, generated by the initial jet instability, impinging on the surface where they determine an unsteady separation-reattachment of the boundary layer. The maximum velocities are encountered at less than 0.05 jet diameters from the surface immediately after the touch down of the main ring-vortex. Due to the unstable separation-reattachment of the boundary layer the flow is Reynolds number (scale) dependent. The maximum in the mean velocity profiles increases and tends towards the wall as the Reynolds number increases. The unsteady Reynolds Averaged Navier–Stokes simulations provide time series which when fitted to the RFD full-scale data, (Gast, K.D., Schroeder, J.L., 2005. Extreme wind events observed in the 2002 thunderstorm outflow experiment. In: Proceedings of the 10th Americas Conference on Wind Engineering (10 ACWE), Baton Rouge, LA, USA), are used to estimate both the length and the velocity scales of the full-scale event. Moreover, these simulations provide a first spatio-temporal flow model for wind loading on structures due to downbursts.

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1. Introduction

1.1. Background

Between 52% and 75% of the extreme annual winds in South-Central United States were registered on thunder days (Twisdale and Vickery, 1992), and 66% of all high intensity wind events in North Eastern United States involving damaging effects on buildings and structures are associated with thunderstorms (based on unpublished recent reports). Also recent analysis of extreme wind speeds in Australia (Holmes, 2002), indicate that thunderstorm downbursts are the extreme wind type at 10 m height.

Field experiments are the most direct method of investigation for High Intensity Winds (HIW) in general and downbursts in particular. The main full-scale downburst studies have been originally initiated in relation to air carrier accidents during take off and landing. Programs such as the Northern Illinois Meteorological Research on Downbursts (NIMROD), the Joint Airport Weather Studies (JAWS), the Classify and Avoid Wind Shear (CLAWS), the Microburst and Severe Thunderstorm (MIST) are essentially meteorological studies oriented towards understanding the formation of microbursts and downbursts, see Alahyari and Longmire (1995) for a complete reference list. It is known that a mass of cold and moist air descends suddenly from the thunderstorm cloud base impinging on the ground surface being afterwards convected radially (Fujita, 1985). According to Hjelmfelt (1988), the diameter of this initial cold air jet varies between $D = 600$ and 1700 m, with the cloud base between $H = 2100$ and 4100 m. The downburst produces intense accelerations of the mass of air in very short periods of time, e.g., from 20 to 100 knots in a couple of minutes, (e.g. the full-scale downburst data Daily Dines, 20/11/59, Australia or the Andrews AFB microburst, Fujita, 1985).

However, a full-scale quantitative spatial and temporal characterization of the flow field generated by these winds mostly for the near ground region (of interest for wind engineering applications) is still lacking. Only recently, detailed field experiments have been conducted to obtain high time resolution data of thunderstorm outflows (Gast and Schroeder, 2005). Hopefully by making simultaneous observations from mobile radars deployed in an organized manner three-dimensional mean flow fields will be shortly available (Schroeder et al., 2005).

Given the complexity of the full-scale phenomena, downburst laboratory simulations have been directed so far towards generic experiments of density currents impinging on a wall (Alahyari and Longmire, 1995, Lundgren et al., 1992) or impinging jets (Bakke, 1957, Cooper et al., 1993, Didden and Ho, 1985, Donaldson and Snedeker, 1971, Landreth and Adrian, 1990, Poreh et al., 1967). Flow visualizations or single point measurements were employed in most cases and the experiments were limited by the laboratory scale. Wind engineering oriented tests were also performed to investigate the pressure distributions on small scale cubic models in stationary and translational downburst-like impinging jets (Chay and Letchford, 2002; Letchford and Chay, 2002).

Numerical simulations of downdrafts (Proctor, 1988), impinging jets (Craft et al., 1993) and mean shear wall jets (Wood et al., 2001) have also been performed but they are essentially meteorological type simulations. While these models take into account thermal stratifications and precipitations they usually make use of rather coarse grids (20 m or larger) and therefore do not emphasize on the near-ground flow dynamics, which is of interest for wind engineering. Also engineering numerical modeling of outdrafts from

thunderstorm events has been attempted by using wall jet velocity profiles and K -epsilon turbulence models (Selvam and Holmes, 1992).

Moreover, some theoretical models (Glauert, 1956; Ivan, 1986; Zhu and Etkin, 1985), as well as engineering usage models (Holmes and Oliver, 2000; Wood et al., 2001) have been also proposed.

1.2. Flow characteristics

The generic case of an impinging jet is an unsteady flow where the vortex rings formed due to the initial Kelvin–Helmholtz instability, impinge on the wall and convect radially afterwards. These main ring vortices trigger an unsteady separation of the wall boundary layer forming a separation reattachment secondary vortex that travels radially preceding the vortex rings dissipation. The intense local stretching produced by these two counter-rotating vortices determines the occurrence of high velocities in the very close to the wall region. It will be shown in Section 3 that the maximum mean velocities are encountered at heights of less than 5% of the initial jet diameter. Therefore for a typical microburst with a diameter of the order of 1000 m the maximum velocity occurs at less than 50 m height, the wind profile being substantially different from the boundary layer (synoptic) wind profiles presently simulated in wind engineering wind tunnels.

Fig. 1 portrays this difference between downburst and synoptic wind mean velocity profiles. Semi-empirical results based on full-scale data (Wood et al., 2001), experimental (Didden and Ho, 1985; Donaldson and Snedeker, 1971), and present numerical results, as well as a generic 1/7 power law boundary layer profile, are plotted for comparison and to better illustrate the important differences between the two wind types. The graph uses self-similar coordinates with the height being non-dimensionalized by the half-velocity height ($z_{0.5 V_{\max}}$).

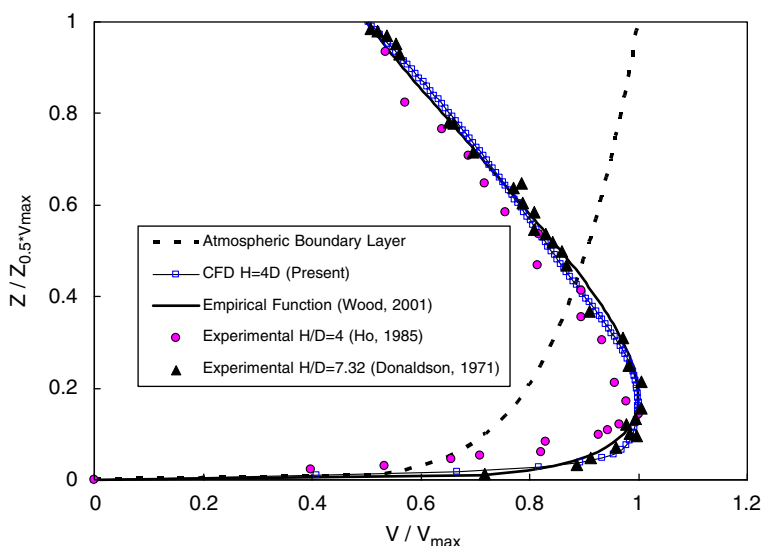


Fig. 1. Downburst mean velocity profile: comparison with laboratory experiments, empirical models and with a typical boundary layer (1/7-power law) profile.

Fig. 2, reproduced from NOAA, presents snapshots of the passage of a real downburst in the proximity of a transmission tower. While the exact perspective of the movie frame picture is not available, the radial passage of a vortex ring is clearly depicted providing qualitative scaling between the vortex, the transmission tower and the height of the cloud base above ground. A region of intense velocity (marked by increased entrainment of dust and/or debris) forms at the forefront of the vortex, very close to the ground. It is obvious that the structural loading derived from such winds would differ radically from a boundary layer wind loading as the maximum wind loading is positioned closer to the ground and the wind mean shear is different.

Moreover, there is a strong possibility for the impinging jet being a Reynolds number (or scale) dependent flow. Its scale dependency can be related to the unsteady separation and reattachment of the boundary layer. An upper bound of this Reynolds number dependency has yet to be established.



Fig. 2. Full scale downburst snapshots; NOAA: <http://www.photolib.noaa.gov/nssl/images/big/nssl0106.jpg>.

The fact that the radial (along the wall) velocity maximum occurs very close to the surface complicates the measurement task: in order to produce high resolution velocity measurements this close to the wall laboratory-based impinging jets have to be relatively large: in order to obtain at least a couple of measurement points in the wall-jet region, the jet diameter has to be of the order of 0.2 m or larger. However, for jet diameters of this size laboratory confinement effects become important (Hangan et al., 2004).

Numerical simulations can overcome some of these difficulties with the additional advantage of providing a characterization of the macro-flow dynamics for the entire field. Experiments, while precise, are most of the time restricted to single or multiple point measurements only. Herein, we attempt to numerically simulate the time-dependent evolution of impinging jets for various Reynolds numbers. We use both steady and unsteady (time-dependent) Reynolds Averaged Navier–Stokes (RANS) simulations. The main motivations for our work are: (i) to better understand the macro-scale dynamics of this type of flow and to explore its time evolution; (ii) to address the Reynolds number dependency problem; (iii) to fit existing full-scale data and determine the length and velocity scales of downbursts in order to generate downburst statistics; and (iv) to generate a first space-time numerical model for further parametric studies, as well as for coupling with time-dependent structural analysis.

2. Numerical simulations

2.1. Numerical set-up

A commercially available Computational Fluid Dynamics (CFD) software, Fluent 6.0 (FLUENT, 2001), was employed to simulate the generic downburst/impinging jet flow. While three-dimensional (3D) vortex dynamics can be directly simulated for all scales using Direct Numerical Simulations (DNS) or over a range of scales using Large Eddy Simulations (LES), these simulations involve very high computational costs mostly for the large Reynolds numbers cases investigated herein. The objective of the current study is to investigate the macro scale flow dynamics of an impinging jet as a generic case for downburst wind flow phenomena. Therefore the time-filtered RANS equations were solved. Considering the RANS time filtering and the computational costs related to the investigation of an increasing range of Reynolds numbers, a 2D-axisymmetric approach was chosen. While this approach does not provide insight into the jet azimuthal instabilities producing secondary precession modes (Hall et al., 2003), it is a robust and economic model allowing for higher Reynolds number simulations.

RANS equations were solved on a structured grid comprising 77,496 (for Reynolds number $Re = 20,000$), 154,104 ($Re = 200,000$) and 416,760 ($Re = 2,000,000$) cells for steady-state simulation and 86,000 ($Re = 20,000$) and 88,019 ($Re = 100,000$) cells for unsteady simulations. The Reynolds number is defined as $Re = V_{\text{jet}} D/\nu$, where D is jet diameter, V_{jet} is jet velocity, constant across the jet outlet and ν is a kinematic viscosity. While the grid size for the unsteady simulations was only marginally increased with Reynolds number, a maximum wall $z^+ = u^*z/\nu$ of less than 1.5 for the first grid points normal to the wall was used in the wall region. The depth of the surface layer corresponding to the maximum radial velocity varies between $z(u_{\text{max}}) = 0.2$ and $0.1D$ for the Reynolds number range investigated herein (Xu and Hangan, 2005). For the extent of the surface layer the non-dimensional cell size based Reynolds number $\Delta^+ = \Delta z u^*/\nu < 5$.

This ensured that the current mesh resolution in the surface region, which is critical for this flow, was adequate. K -epsilon viscosity models are known to over-predict the turbulence kinetic energy near the stagnation point (Cooper et al., 1993). The implications are increased entrainment rates of free-stream fluid followed by a thickening of the radial mean velocity profile. Therefore a Reynolds stress model (RSM), which takes into consideration multiscale and anisotropic effects, was employed herein. Wall reflection terms were included. The default model constants used in the RSM were as follows: $C_{\mu} = 0.09$; $C1$ -epsilon = 1.44; $C2$ -epsilon = 1.92; $C1$ -ps = 1.8; $C2$ -ps = 0.6; $C1'$ -ps = 0.5, $C2'$ -ps = 0.3, turbulent kinetic energy (TKE) Prandtl number = 1; turbulent dissipation rate (TDR) Prandtl number = 1.3. The value of these constants are based on work of Launder and co-workers (Launder, 1989).

For the steady-state simulations, the continuity and momentum equations were solved simultaneously and a spatial second order upwind scheme was employed for flow and a Quadratic Upstream Interpolation for Convective Kinematics (QUICK) scheme for TKE, TDR and for the Reynolds stresses.

For unsteady-state simulations, a segregated approach was used for the solution of the governing equations, and a second order implicit formulation was used for the temporal discretization, with a Semi Implicit Method for Pressure Linked Equation Consistent (SIMPLEC) scheme for pressure–velocity coupling and a QUICK scheme for momentum, TKE, TDR and for the Reynolds stresses. For these temporal and spatial discretization schemes, the SIMPLEC scheme typically provides better convergence (lower convergence residuals) compared to a pressure-implicit with splitting of operators (PISO) scheme.

All grids were structured grids with the wall region strenuously refined. Near wall regions were modeled using an enhanced wall treatment (FLUENT, 2001) that is a near-wall modeling method combining a two-layer model with enhanced wall functions. If the near-wall mesh is fine enough to be able to resolve the laminar sub layer, then the enhanced wall treatment will be identical to the traditional two-layer zonal model. Grid dependency tests were performed for the steady-state simulations only, the grid size being increased in 25% increments until no noticeable variance in the mean velocity profiles was achieved.

Fig. 3 shows the schematic diagram of the computational domain. X -axis corresponds to the radial direction and Z -axis corresponds to the axial direction, respectively. The grid is not shown because due to its intense refinement the entire wall region becomes shaded.

The size of the two-dimensional computational domain was $(7 \times 9 D)$ for the steady-state simulations and $(10 \times 10 D)$ for the unsteady simulations. For the unsteady simulations, the mesh resolution in the wall region was maintained at the same level with steady-state simulations (for same Reynolds numbers) while the computational domain was increased in order to reduce the effects of the convective vortex approaching the outlet boundary. The distance between the jet and the surface was $H/D = 2$ and 4 for the steady-state simulations and $H/D = 4$ for the unsteady simulations. These H/D values were chosen for benchmarking against two of the available detailed laboratory experiments (Cooper et al., 1993; Didden and Ho, 1985). It is expected that this is not a strong parameter as long as the distance between the jet and the surface, H , allows for the formation of the main vortex rings (Hangan et al., 2004). The boundary conditions are summarized in Table 1 and the solution control parameters are specified in Table 2.

For the steady-state simulations three jet diameters were considered $D = 0.0381$, 0.381 and 3.81 m corresponding to Reynolds numbers of 20,000, 200,000 and 2,000,000, respectively.

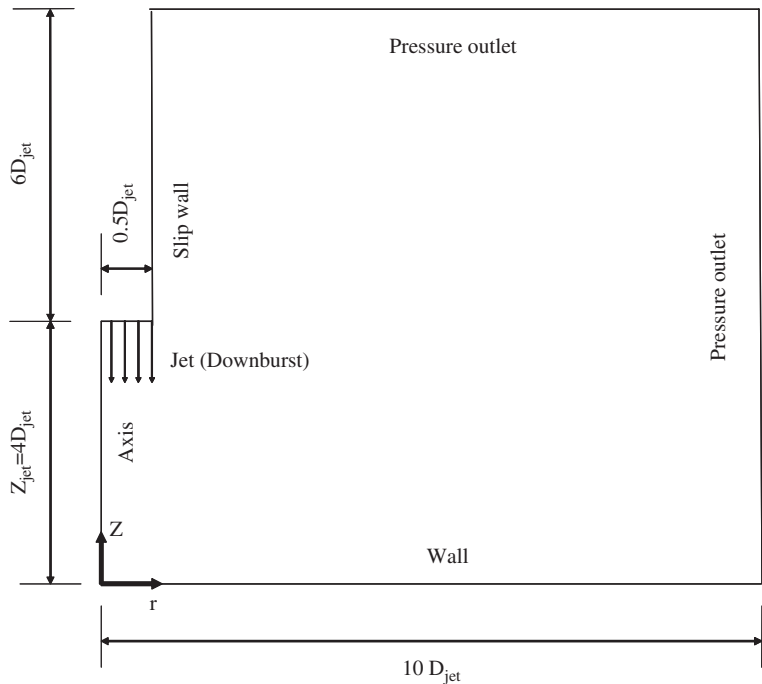


Fig. 3. Schematic diagram of computational domain.

Table 1
Boundary conditions for unsteady state

Jet (inlet velocity)	$V_{in}/V_{jet} = 1$ Turbulence intensity = 1% Hydraulic diameter = D
Axis	Axis
Wall	Enhanced wall treatment
Pressure outlet	Backflow turbulence intensity = 1% Backflow hydraulic diameter = D
Slip wall	Symmetry

Table 2
Solution control parameters: under relaxation factors for unsteady state simulation

Pressure	0.3
Momentum	0.7
Turbulent kinetic energy (TKE)	0.8
Turbulent dissipation rate (TDR)	0.8
Turbulent viscosity	1
Reynolds stresses	0.5

The time dependent (unsteady) simulations were performed for $Re = 20,000$ as well as for $Re = 100,000$. A non-dimensional time step $dt^* (= dt V_0/D) = 0.001$ was used for both Reynolds numbers generating approximately 3400 (for $Re = 20,000$) and 3300

(for $Re = 100,000$) time steps per shedding period ($1/f_s$) corresponding to $St = 0.294$ for $Re = 20,000$ and $St = 0.304$ for $Re = 100,000$, where St is the Strouhal number defined as $St = Df_s/V_{jet}$. The frequency, f_s , associated with the “shedding” of the ring vortices due to the initial Kelvin–Helmholtz jet instability was determined based on the time interval between the first peak and the second peak of the radial velocity time history data measured at the position $r/D = 1$ and $z/D = 0.02$. The above time resolutions were considered satisfactory relative to the time scales to be resolved.

2.2. Steady-state simulations

Steady state, axysymmetric simulations were conducted for Reynolds numbers between 20,000 and 2,000,000. The numerical results compare well with the mean radial velocity profiles from experiments (Hangan and Xu, 2005) for $H/D = 4$ and $Re = 20,000$ (numerical) and 27,000 (experimental) for radial distances, r/D , between 1 and 2.5, Fig. 4a. Benchmarking is also provided in Fig. 4b, where experimental (Cooper et al., 1993; Hangan et al., 2004) and previous workshop K -epsilon numerical results (Brison and Brun, 1991) are compared with present numerical results for $H/D = 2$ and $r/D = 2.5$. For

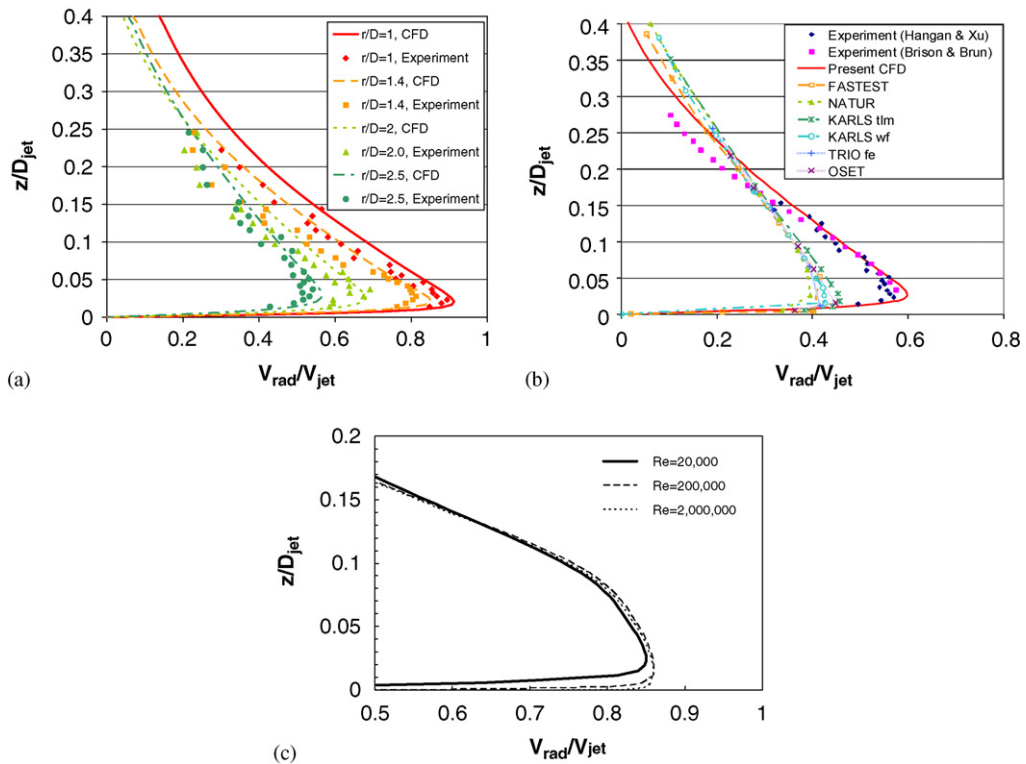


Fig. 4. (a) Comparison of present numerical results with experiments (Hangan and Xu, 2005) for $H/D = 4$ and $Re = 20,000$ (numerical) and 27,000 (experimental). (b) Comparison of past (Brison and Brun, 1991) and present numerical results with experiments (Cooper et al., 1993; Hangan and Xu, 2005), for $H/D = 2$, $r/D = 2.5$ and $Re = 20,000$ (numerical and Cooper et al., 1993) and (c) Reynolds number dependency, present numerical results $H/D = 4$, $r/D = 1$.

benchmarking purposes the results are normalized by the jet diameter and velocity. The K -epsilon numerical simulations (Brison and Brun, 1991) produce thicker profiles compared to experiments. This thickening corresponds to a higher entrainment rate of the outer fluid, a consequence of over-predicting turbulent energy at impingement. In contrast, the present results compare very well to experiments.

When results for various Reynolds numbers are compared, the steady-state numerical results suggest that the mean velocity profile changes with scale, see Fig. 4c, in formal agreement with experimental results (Hangan and Xu, 2005; Zhu and Etkin, 1985). Notably it appears that the height above the surface corresponding to the maximum velocity decreases with increasing Reynolds number while the maximum velocity decreases. This Reynolds number dependency appears to be more pronounced for $Re < 200,000$.

Reynolds number dependency is frequently associated with flow unsteady separation/reattachment. For impinging jets this separation-reattachment relates to the temporal evolution of the flow macro-scales and therefore unsteady simulations were further conducted.

2.3. Time-dependent simulations

Unsteady (time-dependent) simulations were conducted for $H/D = 4$ and $Re = 20,000$ and 100,000 in order to better understand the macro-scale dynamics of the flow. The results of these simulations show that as the jet flow is impulsively started a strong Kelvin–Helmholtz instability forms due to the shear between the jet flow and the ambient, still air. The result of this instability is the formation of ring vortices, which are then convected towards the impinging surface. Figs. 5 and 6 ($Re = 20,000$ and 100,000, respectively) present snapshots of the flow field in terms of vorticity contours and velocity vectors for various non-dimensional time steps: $T = dt^*$ ($= dt V_{jet}/D_{jet}$). The six snapshots correspond to moments when: (a) the ring-vortex approaches the surface; (b) and (c) the vortex touches down and triggers the unsteady separation-reattachment of the boundary layer; (d) the main ring-vortex and the counter-rotating vortex formed inside the reattachment bubble travel radially; (e) and (f) a second ring vortex (the results of the initial quasi-periodic Kelvin–Helmholtz instability) touches down and the two ring vortices travel radially and dissipate. While the formation of the separation-reattachment secondary vortex might slow down the radial advection of the ring vortex (Alahyari and Longmire, 1995), the two counter-rotating vortices produce a region of intense stretching in between them. This stretching translates into an important local flow acceleration in the immediate vicinity of the wall with maximum time averaged velocities manifested around the radial position $r/D = 1.1$. This maximum velocity is of the same order of magnitude as the jet outflow velocity, but varies with Reynolds number, in agreement with the steady-state results presented in Fig. 4.

The time evolution of the radial and axial velocity profiles for $Re = 20,000$ are presented in Figs. 7 and 8, respectively, at various radial positions: (a) $r/D = 0.5$; (b) $r/D = 1$; (c) $r/D = 1.5$ and (d) $r/D = 2$ for four of the time steps T corresponding to Fig. 5. The maximum radial velocity is reached when the ring vortex touches down at $T = 9.8$, Fig. 7c. The axial velocity reaches fairly high levels, of the order of 0.6 of the radial component, Fig. 8d. However, the axial and the radial maxima are not reached at the same time or same radial position. The relative high axial (vertical in a downburst) velocities constitute

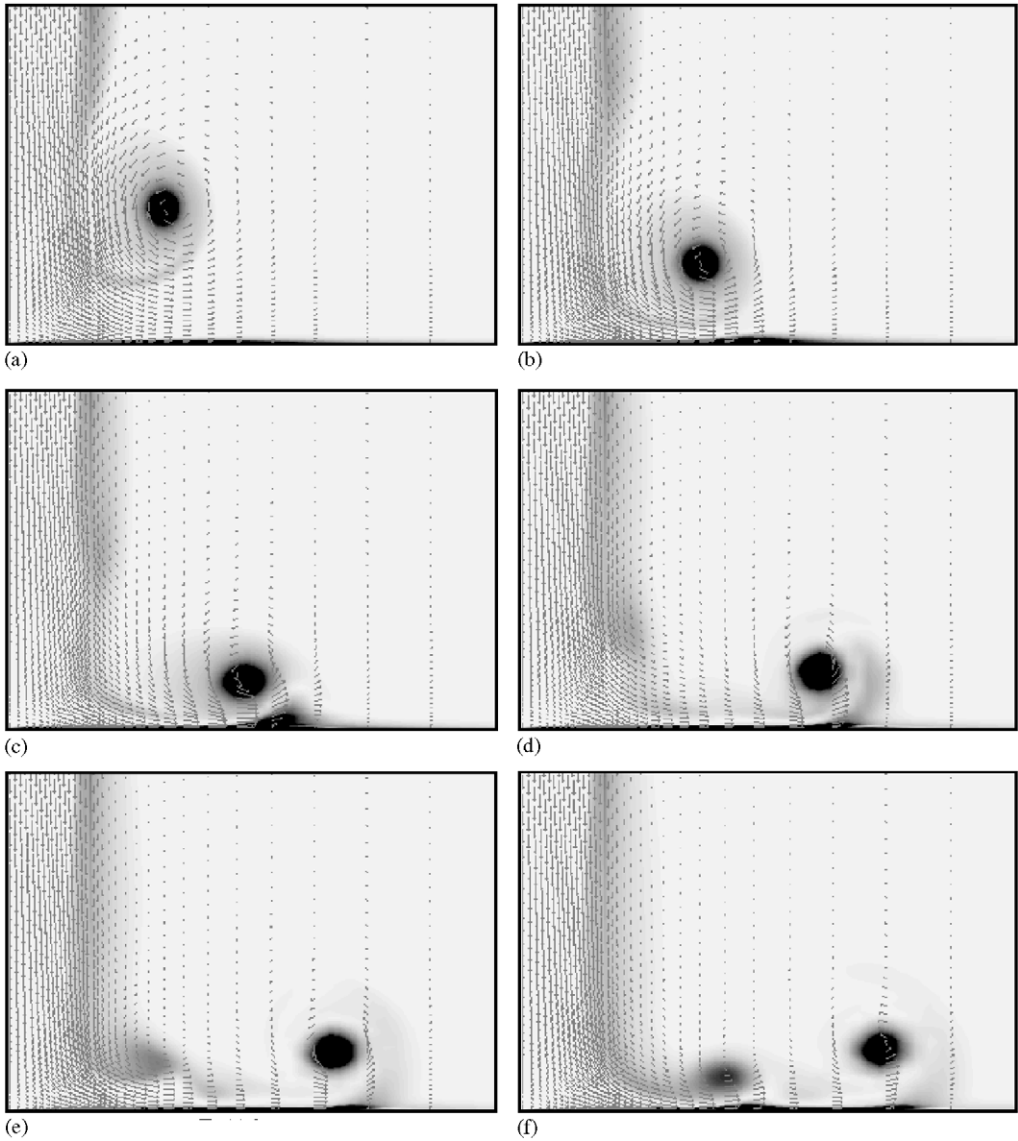


Fig. 5. Vorticity and vector snapshots; vortex evolution ($Re = 20,000$) (a) $T = 7.8$; (b) $T = 8.8$; (c) $T = 9.8$; (d) $T = 10.8$; (e) $T = 11.8$; and (f) $T = 12.8$.

yet another important factor that differentiates downburst generated winds from synoptic, boundary layer winds.

When comparing the time-dependent profile plots for the two Reynolds numbers, $Re = 20,000$ and $100,000$ (not shown herein) one may infer that the maximum velocity vector is less inclined (the axial component is relatively smaller) for the $Re = 100,000$ when compared to $Re = 20,000$. As the maximum velocity vector corresponds to the stretching region between the two counter-rotating vortices the decreasing inclination with increasing

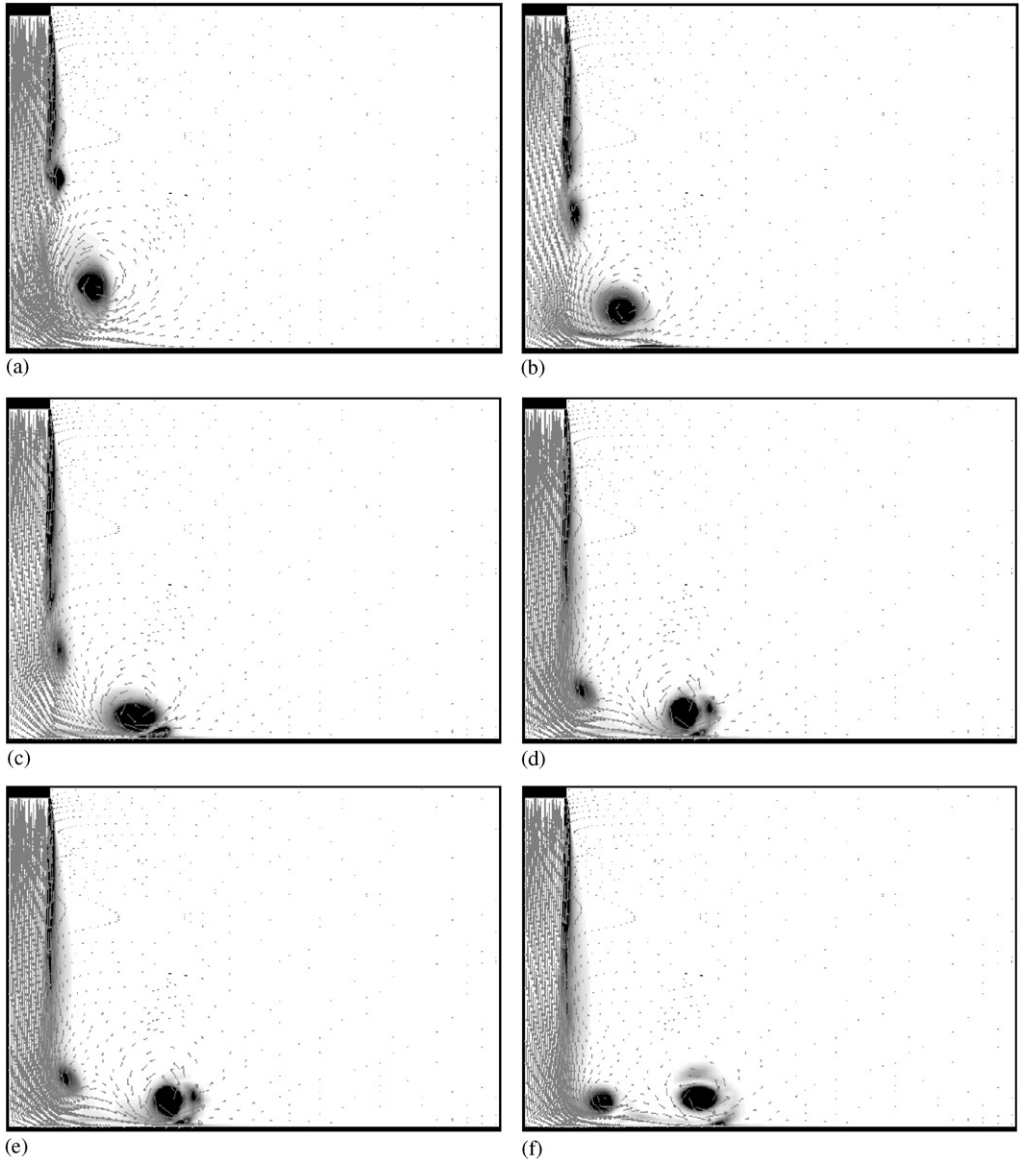


Fig. 6. Vorticity and vector snapshots; ring-vortex and secondary vortex evolution ($Re = 100,000$). (a) $T = 7.8$; (b) $T = 8.8$; (c) $T = 9.8$; (d) $T = 10.8$; (e) $T = 11.8$; and (f) $T = 12.8$.

Reynolds number may be associated with a relative decrease in boundary layer depth. It follows that, for smooth surfaces, the flow would asymptotically tend towards a quasi-inviscid behavior as the Reynolds number (or scale) increases. It is expected that at these high Reynolds numbers surface roughness would act inversely, increasing the boundary layer depth (Hangan and Xu, 2005). Also when the velocity–vorticity snapshots for the two Reynolds numbers are compared, Figs. 5 and 6, the flow is found to be more periodic for

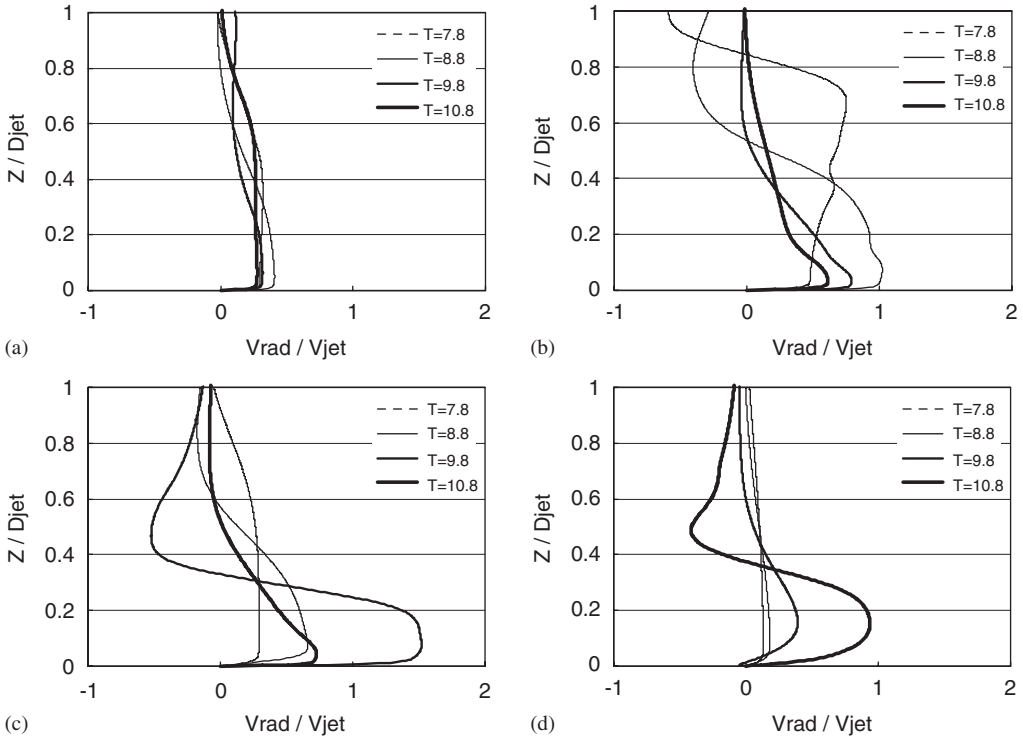


Fig. 7. Radial velocity evolution with time: $Re = 20,000$ (a) $r/D = 0.5$; (b) $r/D = 1.0$; (c) $r/D = 1.5$; and (d) $r/D = 2.0$.

the higher Reynolds number. This flow behavior is consistent with surface pressure measurements results (Hangan et al., 2004). At low Reynolds numbers ($Re \leq 20,000$) the initial Kelvin–Helmholtz instability is weak and the flow is weakly periodic. As the Reynolds number increases ($Re > 20,000$) the instability intensifies and the flow becomes increasingly organized and periodic.

Present numerical results are compared with a semi-empirical model proposed by Holmes and Oliver (2000) in Fig. 9. The comparison is based on a microburst diameter of 500 m. This figure suggests that Holmes and Oliver model reproduces quite well the steady-state radial flow field at 10 m height. Meanwhile, the numerical time-dependent radial profiles show a pronounced variability with negative values (immediately after the ring-vortex touch down) and with secondary peaks (corresponding to the passage of consecutive ring vortices).

Fig. 10 depicts the time evolution of the maximum radial velocity and of the height at which this maximum velocity occurs at $r/D = 1.5$. The left-hand side vertical axis presents the maximum radial velocity divided by the jet velocity while the right-hand side axis shows the height at which the maximum occurs divided by the jet diameter. After a transitory period ($T < 8$) the maximum radial velocity shows two extreme values at $T = 10$ and 13 corresponding to the passage of the first two ring vortices. The extremes occur at the time of the vortex touch down and correspond to a height of less than 5% of the initial jet diameter ($0.05 D$).

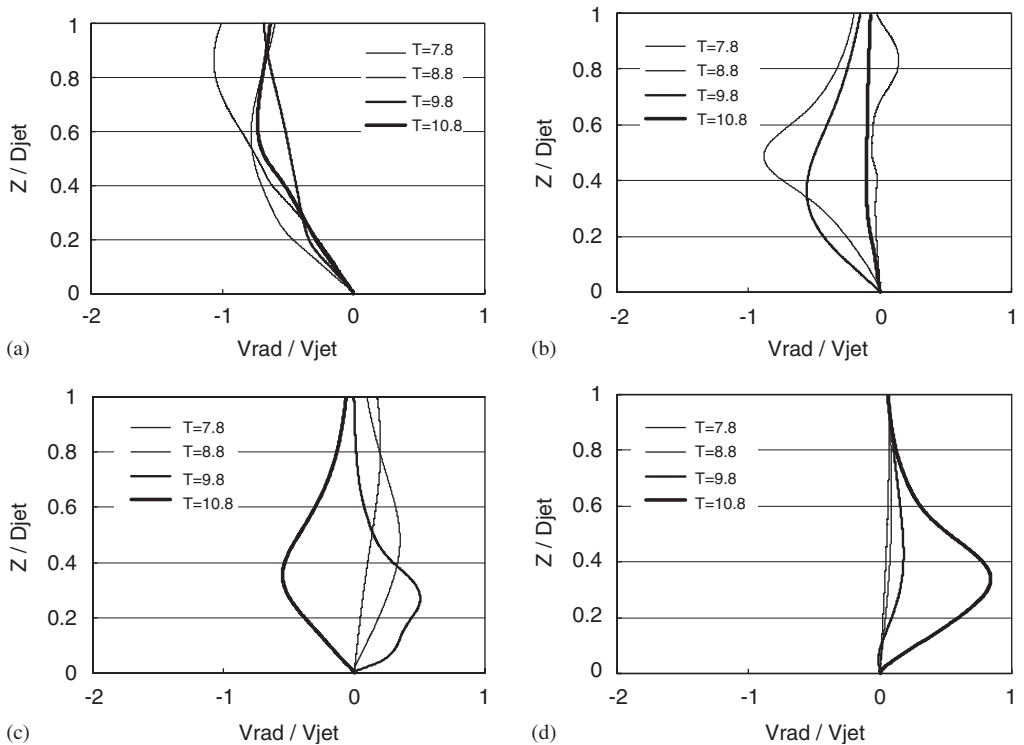


Fig. 8. Axial velocity evolution with time: $Re = 20,000$. (a) $r/D = 0.5$; (b) $r/D = 1.0$; (c) $r/D = 1.5$; and (d) $r/D = 2.0$.

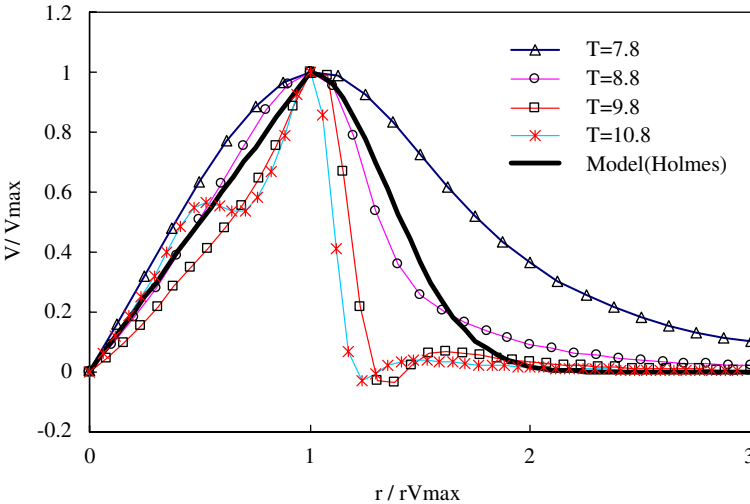


Fig. 9. Radial velocity evolution at $z/D = 0.02$: Numerical time dependent results ($Re = 20,000$) are compared to Holmes and Olivier's model (2000).

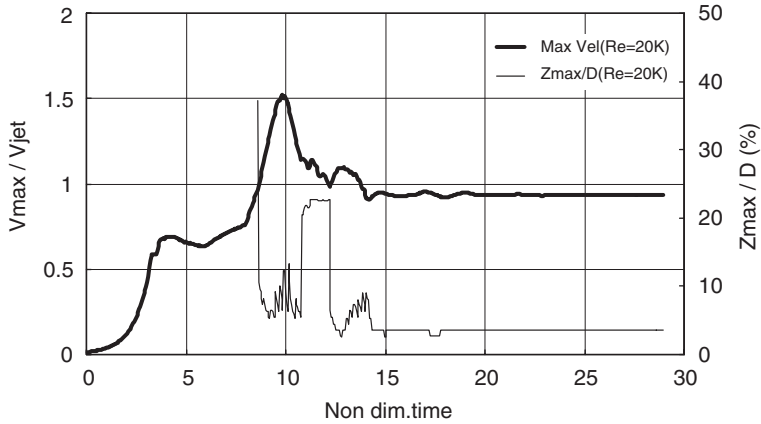


Fig. 10. Maximum Radial velocity and the corresponding height ($z_{\max}$) evolution with time at $r/D = 1.5$.

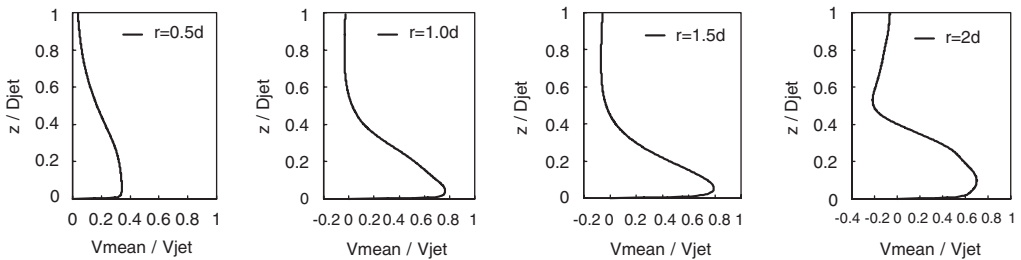


Fig. 11. Time averaged velocity profiles measured at $r/D = 0.5, 1.0, 1.5, 2.0$. The average as taken over one shedding period. $Re = 20,000$.

Time averaged velocity profiles can be calculated at every radial position (r/D), see Fig. 11, for instance by averaging the time history data over one period of the jet Kelvin–Helmholtz instability, between two radial velocity consecutive maxima.

Maximum radial velocity profiles ($Re = 20,000$) are presented in Fig. 12. These values are the maximum values extracted from the entire time series at every point in the domain and therefore they may not occur at the same moment in time. Also, given the RANS filtering approach, these values should not be regarded as peak values but rather maximum large-scale (ensemble averaged) values. The profiles are plotted at the same radial distances as the time-averaged velocity profiles (over one ring-vortex period) in Fig. 11.

2.4. Comparison with full-scale downburst data

Fig. 13 compares the results from the present unsteady RANS simulations with full-scale velocity time series from the rear-front downburst (RFD) data from Tower 4, (Gast and Schroeder, 2005).

The full scale data corresponds to a translating downburst event as indicated by the sudden 180° change in wind direction occurring at time $T \cong 675$ s, in Fig. 4 from Gast and

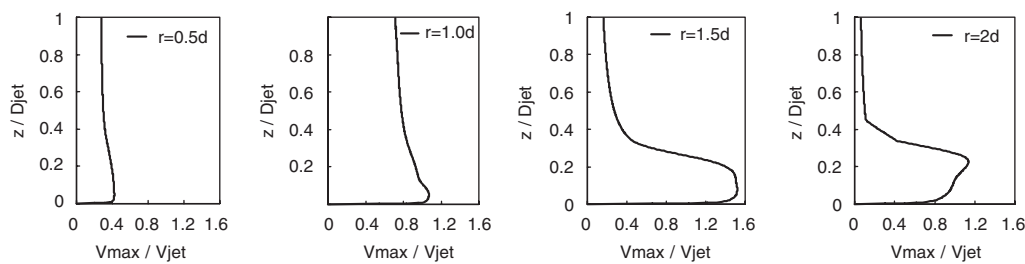


Fig. 12. Max velocity profiles at $r/D_{jet} = 0.5, 1.0, 1.5, 2.0$: at every $(r/D, z/D)$ point, max. radial velocity is measured based on the entire time.

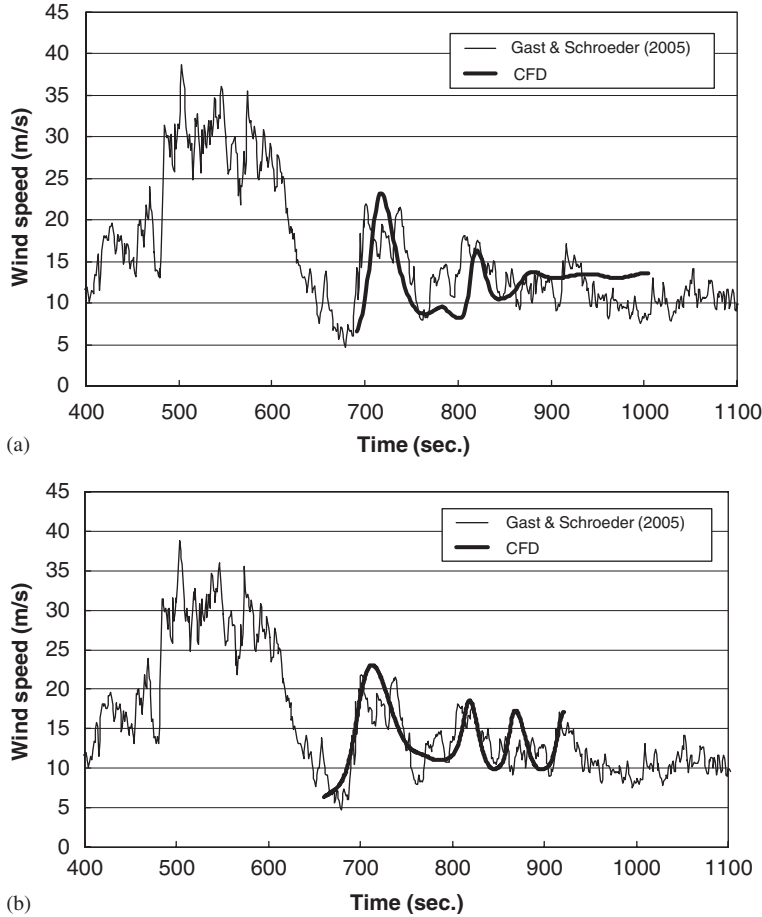


Fig. 13. Comparison of CFD results ($Re = 20,000$) with the RFD event (Tower 4) full-scale velocity time series: (a) $H/D = 4$; $D = 600$ m, $U_t = 6$ m/s; and (b) $H/D = 2$, $D = 950$ m, $U_t = 6$ m/s.

Schroeder (2005). Therefore in order to compare the two data sets the translation velocity, U_t , of the full-scale event needs to be estimated.

If it is assumed that Tower 4 is positioned close to the center of the passing downburst, based on the CFD analysis, the large velocity peaks at $T \cong 500, 540$ and 580 s would correspond to the passage of ring vortices in the direction of the downburst translation. Therefore their magnitude of approx. 35 m/s corresponds to the sum of the translation velocity and the maximum velocity in a stationary impinging jet: $U_t + V_{jet}$. The lower velocity peaks at $T \cong 725, 825$ and 925 correspond to the passage of ring vortices on the opposite side of the downburst and therefore the first peak at 725 s would correspond to: $U_t - V_{jet} \cong 23$ m/s. The difference between the wind speed magnitude of the high peak at $T = 580$ s and the peak at $T = 725$ s gives twice the translation velocity of the downburst: $(U_t + U_{jet}) - (U_{jet} + U_t) = 2U_t = 35 - 23 = 12$. Therefore the downburst translation velocity is

$$U_t \cong 6 \text{ m/s.} \quad (1)$$

Moreover, if the minimum wind speed at $T = 675$ s is interpreted as the passage of the downburst centre (roughly corresponding to the change in direction) that wind speed should also approximate the translation downburst velocity, in excellent agreement with value determined above. It follows immediately that the corresponding jet velocity is

$$V_{jet} \cong 35 - U_t = 29 \text{ m/s.} \quad (2)$$

The two consecutive peaks at $T = 725$ and 825 s are interpreted as the passage of two consecutive ring vortices. The time lag in between these two vortices in the translating downburst is therefore $T_{st} = 100$ s. Also, the time lag between two passing vortices in a stationary impinging jet, T_s , can be expressed based on the initial jet Strouhal number (St) as

$$T_s = D / (St V_{jet}), \quad (3)$$

where D and V_{jet} are the diameter and the initial impinging jet velocity. The relative distance, λ , between two consecutive vortices is assumed to be the same for a stationary and a translating jet and can be expressed as a function of the vortex convective velocity, U_c , as

$$\lambda = U_c T_s = (U_c \pm U_t) T_{st}, \quad (4)$$

where the $\pm$ sign corresponds to vortices being convected in the same or opposite direction to the downburst translation. Taking $U_c / V_{jet} = 0.65$ (see Landreth and Adrian, 1990) and using relation (3) and (4) the equivalent downburst jet diameter can be expressed as

$$D = [(U_c \pm U_t) St T_{st}] V_{jet} / U_c, \quad (5)$$

where T_{st} is the time difference between two consecutive velocity peaks with: $T_{st} \cong 825 - 725 = 100$ s for the peaks corresponding to the ring vortices opposing translation and $T_{st} \cong 540 - 500 = 40$ s for the peaks on the translation side.

For a free jet the Strouhal number is a function of the axial distance (Popiel and Trass, 1991):

$$St = 1.2(x/D)^{-1}. \quad (6)$$

Therefore for jet to surface distances $H/D = 4$ and $H/D = 2$ the corresponding Strouhal numbers near impingement are $St = 0.3$ and 0.6 , respectively. The present numerical unsteady simulations produced results which are in excellent agreement with

relation (6): For $H/D = 4$, $St = 0.294$ at $r/D = 1$ and $St = 0.29$ at $r/D = 1.5$ were obtained. For $H/D = 2$, $St = 0.63$ was obtained (Kim et al., 2005). Using relation (5) the equivalent jet diameter is found to be: $D \cong 600$ and 1200 m for $H/D = 4$ and 2 , respectively.

The present CFD generated and RANS filtered time series reproduce the passage of consecutive (and quasi-periodic) vortex rings generated by a stationary impinging jet (generic downburst). Therefore the CFD data is fitted with RFD full scale data (Gast and Schroeder, 2005), for $T > 675$ s aiming at reproducing the same periodicity as in full scale. Based on the above arguments in matching the two sets of data the downburst translation velocity $V_t = 6$ m/s is used. The full-scale data from Tower 4 at 10 m height was fitted with the CFD data corresponding to $H/D = 4$ and 2 at position $(r/D, z/D) = (1.5, 0.02)$, Figs. 13a and b, respectively. Based on the length and velocity scaling, the best fit of the CFD data was obtained for $V_{jet} = 29$ m/s and for downburst initial diameters $D = 600$ and 950 m corresponding to $H/D = 4$ and 2 . These values are in good agreement with the values obtained from the full-scale data analysis.

3. Discussions and conclusions

The numerical results presented herein help better understanding the macro-flow dynamics of impinging jets with applications to downburst simulations. The main findings are summarized below:

- (i) *Time-space dependency*: Impinging jets, the primary generic flows for the simulation of downburst winds, are time and space dependent.

In the free-jet region the initial Kelvin–Helmholtz instability generates quasi-periodic vortex rings. The first formed vortex (immediately after the impulse starting) is the most intense. This fact is attributed to the stronger shear generated at start-up when the surrounding fluid is at rest compared to the steady-state situation (usually simulated in laboratory) where the surrounding fluid has been already entrained in the axial direction by the jet flow. The Strouhal number, $St = f_s D / V_{jet}$, associated with these vortex rings varies with the distance between the initial jet and the surface with values between 0.3 (for $H/D_{jet} = 4$) and 0.6 (for $H/D_{jet} = 2$). The Strouhal number determined by the present unsteady numerical simulation fits closely previous (Popiel and Trass, 1991) experimental results. A full scale generic downburst of 2000 m in diameter and with an initial velocity, $V_{jet} = 50$ m/s would have a characteristic ring vortex frequency of the order of 0.01 Hz which is well below the natural frequency of structures with the exception of very elastic structure systems such as transmission lines.

An important question related to the time dependency is whether after filtering out the large-scale flow component, the remaining component is random and statistically similar to the one present in boundary layers. The results of recent research tend to be dependent on the choice of the filtering method (Chen and Letchford, 2005; Gast and Schroeder, 2005; Wang and Kareem, 2005).

In the impinging jet region the vortex rings touch the surface and start convecting radially triggering an unsteady separation of the newly formed boundary layer. The presence of this system of counter-rotating, radially traveling vortices determines time-dependent velocity profiles with the maximum velocity registered very close to the surface ($z_{max} \leq 0.05 D$) and at radial distances from the jet axis of $r_{max} = 1.1–1.5 D$.

Present unsteady numerical simulation fits very closely to previous experimental data. In the wall jet region (approx. $r > 1.4D$) the main ring vortices gradually dissipate as they travel radially. The flow reaches local equilibrium and the mean velocity profiles become self-similar. This is the region of the flow that can be easily modeled either numerically (Sengupta et al., 2001) or analytically (Xu and Hangan, 2005).

Overall the flow field associated with impinging jets (and therefore downbursts) is both spatially and temporally different from a boundary layer flow (and therefore from synoptic winds). Fig. 14 underlines this spatio-temporal variability of the phenomenon by showing the time variation of the radial velocity at $r/D = 1$ at various heights, for $Re = 20,000$. This particular time and space localization will produce different load-shapes compared to the ones produced by boundary layer flows. Moreover, the structural loading would depend on such parameters as the initial downburst diameter, the height above the ground or the position of the downburst relative to the structure (both radially and azimuthally) (Kim et al., 2005).

- (ii) *Scale dependency*: The Reynolds number dependency of this type of flow (primarily attributed to the unsteady separation of the boundary layer) has been observed both in the mean and the unsteady flow field. The maximum velocity in a profile increases and the location of this maximum approaches the surface as the Reynolds number increases. Also the flow becomes more organized (periodic) with increasing Reynolds number. This can be related to the initial jet-outlet shear effects.

The flow dependency with scale raises the issue of correctly simulating generic downburst flows through laboratory experiments. Fortunately, it seems that the scale dependency becomes less pronounced towards higher Reynolds numbers and an asymptotic behavior is expected with the surface roughness becoming the dominant parameter at large scales (Hangan and Xu, 2005).

- (iii) *Comparison with full-scale data*: The understanding of the macro-flow dynamics sheds new light on the interpretation of full-scale data. Comparison between present RANS filtered CFD time series and the 2002 RFD outflow experiment (Schroeder et al., 2005) shows good agreement and provides new means in relating full-scale time-histories to the flow dynamics. It is shown how the translation velocity of a downburst can be estimated and how, by fitting full scale data with numerical data, the length and the

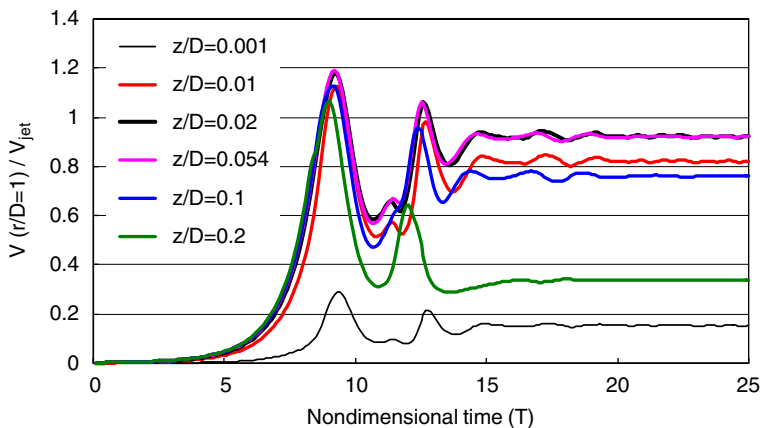


Fig. 14. Normalized radial velocity measured at $r/D = 1$ and various heights ($Re = 20,000$).

velocity scales can be determined in terms of the initial jet diameter, D , and the initial jet velocity, V_{jet} . This CFD-based scalling of full-scale data can be further used to generate downburst statistics.

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